

Chemistry

Unit 1

Practice Exam 2 Answers

Section one: Multiple-choice

Question 1

- D** The mass number is the sum of the number of protons and neutrons. So for this ion the mass number is $11 + 12 = 23$. The ion has 1 fewer electrons than protons so it has a charge of +1.

Question 2

- B** The electron configuration is 2,1. The group number is equal to the number of valence electrons so it would be found in group 1. The period number is the number of shells containing electrons. There are 2 electron shells so it would be found in period 2.

Question 3

- D** The group number is equal to the number of valence electrons; the number of electrons in the outer electron shell.

Question 4

- C** Isotopes have the same atomic number but different mass numbers.

Question 5

- B** X has an atomic number of 19 so 19 p^+ in the nucleus. Its electron configuration is 2,8,8 so it has 18 e^- . Therefore, as it has 1 more p^+ than e^- , X is a positively charged ion.

Question 6

- A** Al has 13 electrons and an electron configuration of 2,8,3. So the Al^{3+} ion will have the electron configuration of 2,8.

Ca has 20 electrons and an electron configuration of 2,8,8,2. So the Ca^{2+} ion will have the electron configuration of 2,8,8.

Question 7

- C** Silicon has a covalent network structure where all electrons are localised. The valence electrons are shared between atoms and lie mostly in the region between the two nuclei.

Question 8

- B** Ionic bonding involves electron transfer not sharing of electrons.

Question 9

- D** P, with 5 valence electrons, will form 3 covalent bonds. Q, with 7 valence electrons, will form 1 covalent bond. So 3 Q atoms will bond with each P atom and the formula is PQ₃.

Question 10

- B** The proportion of N is given by M(N) in compound/M(compound). So, it is in
- a** 14/17
 - b** 28/32 or 14/16
 - c** 14/63
 - d** 28/80 or 14/40

So 14/16 represents the highest proportion of N.

Question 11

- A** It is an alkane with 3 carbons so propane is correct. There is only one way to arrange 3 carbon atoms so there are no isomers. Therefore no number prefix is used.

Question 12

- A** As heat is added for the reaction to proceed, the reaction is said to be endothermic. The energy of reactants is less than that of products so, as ΔH is defined as $H(\text{products}) - H(\text{reactants})$, its sign is positive.

Question 13

- C** The molar mass is the mass, in gram, of 1 mol of substance. So its unit is gram per mol, or gram mol⁻¹.

End of section one

Section two: Short answer

(Note: In multi-mark questions each allocated mark is indicated as * and total marks are indicated in parenthesis.)

Question 14

- a** MgCl_2^*
 $\text{Al}_2(\text{SO}_4)_3^*$
 SnO_2^* (3 marks)
- b** Iron(III) nitrate*
Ammonium chloride* (2 marks)

Question 15

- a** Co. Fe has 26 protons, Co 27 protons. (1 mark)
- b** Li. Electronegativity decreases down a group, so the first element has the highest electronegativity. (1 mark)
- c** F^- . Reactivity of non-metals decreases down a group so F is the most reactive and it forms an ion with a -1 charge. (1 mark)
- d** Mg. 3 occupied shells indicates it is in period 3; 2 valence electrons indicates group 2, hence Mg with electron configuration of 2,8,2. (1 mark)
- e** Any element from Th to Lr. The actinides are the series of 14 elements that follow actinium in the periodic table. (1 mark)
- f** Ne. The noble gases are in group 18. (1 mark)

Question 16

- a** The atomic radius of rubidium is larger*. Rubidium has one extra electron shell occupied*. (2 marks)
- b** The first ionisation energy of sulfur is larger*. Both have the same number of occupied electron shells but sulfur has a greater nuclear charge* so more energy is required to remove an electron from sulfur than magnesium*. (3 marks)
- c** The electrical conductivity of liquid potassium chloride is higher*. Potassium chloride consists of ions which are free to move in the liquid state but propane consists of neutral molecules*. (2 marks)

Question 17

- a** 10^{-9} m (1 mark)
- b** They have a much larger surface area to volume ratio* so surface effects (such as reactions that occur on the surface of catalysts or friction effects) become much more significant*. (2 marks)
- c** Any valid application including catalysis, designer drugs, transparent sunscreens, anti-graffiti coatings for walls, stain-repellent fabrics, self-cleaning windows, ceramic coatings for solar cells, etc. (1 mark)

Question 18

- a** Allotropes are different physical forms of the same element. (1 mark)
- b** In graphite each carbon is bonded to 3 other carbon atoms forming a hexagonal arrangement in layers of covalently bonded atoms*. The fourth electron from each carbon atom is delocalised and is free to move between the layers*.
In diamond each carbon is covalently bonded to 4 other carbon atoms*. The bonds are continuous throughout forming a 3 dimensional lattice*. (4 marks)
- c** The forces between layers of carbon in graphite are very weak so when the 'lead' is pushed across the paper, layers of carbon atoms are left behind. (1 mark)

Question 19

- a** Mass spectrometer (1 mark)
- b** $A_r(\text{Ir}) = \frac{(37.30 \times 190.97) + (62.70 \times 192.97)^*}{100}$
 $= 192.2^*$ (answer to 4 significant figures) (2 marks)
- c** Atomic absorption spectroscopy relies on identifying electron transfer between atomic energy levels. (1 mark)

Question 20

- a** $m(\text{vitamin C}) = 40 \text{ mg} = 0.0400 \text{ g}$
 $n(\text{vitamin C}) = m/M = 0.0400/176.1 = 2.27 \times 10^{-4} \text{ mol}$ (1 mark)
- b** Number of molecules present $= n \times N_A = 2.27 \times 10^{-4} \times 6.022 \times 10^{23}$
 $= 1.37 \times 10^{20}$ (1 mark)
- c** Mass of oxygen in 1 mole of vitamin C $= \frac{54.5 \times 176.1}{100}$
 $= 96.0 \text{ g}^*$
 $n(\text{O}) \text{ in 1 mole of vitamin C} = 96.0/16 = 6$
So each molecule of vitamin C contains 6 atoms of oxygen*. (2 marks)

Question 21

- a**
- | | | | | |
|------------------------|---|----------------------|---|----------------------|
| $n(\text{C})$ | : | $n(\text{H})$ | : | $n(\text{Br})$ |
| $= \frac{12.8}{12.01}$ | : | $\frac{2.13}{1.008}$ | : | $\frac{85.07}{79.9}$ |
| $= 1.07$ | : | 2.11 | : | 1.07^* |
| $= 1$ | : | 2 | : | 1^* |
- (Divide by smallest ratio.)
So EF is CH_2Br^* . (3 marks)
- b** RFF of $(\text{CH}_2\text{Br}) = 93.9$. Since the relative molecular mass is 188, the molecular formula is $\text{C}_2\text{H}_4\text{Br}_2$. (1 mark)

End of section two

Section three: Extended answer

Question 22

- a** Alkanes (1 mark)
- b** Ethane (1 mark)
- c** $\text{C}_2\text{H}_6(\text{g}) + \text{Cl}_2(\text{g}) \rightarrow \text{C}_2\text{H}_5\text{Cl}(\text{g}) + \text{HCl}(\text{g})$ (1 mark)
- d** **i** One C–H bond and one Cl–Cl bond (1 mark)
- ii** Absorbs energy (1 mark)
- e** **i** One C–Cl bond and one H–Cl bond (1 mark)
- ii** Releases energy (1 mark)
- f** If energy is released during the reaction, it means that the energy released when bonds form is greater than the energy required to break bonds; so the bonds formed must be overall stronger than the bonds broken. (1 mark)
- g** **i** The heat of reaction is the amount of heat released or absorbed during a chemical reaction. (1 mark)
- ii** As heat is released during the reaction, by definition, the heat of reaction will have a negative value. (1 mark)
- iii** A reaction that releases heat is exothermic. (1 mark)

Question 23

- a** **i** $m(\text{C})$ used each day = $36\,000 \times 23/100 = 8280$ tonnes = $8.28 \times 10^9 \text{ g}^*$
 $n(\text{C})$ used each day = $\frac{8.28 \times 10^9}{12.01} = 6.89 \times 10^8 \text{ mol}^*$ (2 marks)
- ii** $n(\text{CO}_2)$ formed = $n(\text{C})$ used = $6.89 \times 10^8 \text{ mol}^*$
 $m(\text{CO}_2)$ formed = $n(\text{CO}_2) \times M(\text{CO}_2)$
= $6.89 \times 10^8 \times 44.01 = 3.03 \times 10^{10} \text{ g}$ (or $3.03 \times 10^4 \text{ t}$)^{*} (2 marks)
- iii** $m(\text{O}_2)$ = $10.0 \text{ kg} = 10\,000 \text{ g}$
 $n(\text{O}_2)$ = $10\,000/32.00 = 312.5 \text{ mol}^*$
 $n(\text{C})$ needed = 312.5 mol
 $m(\text{C})$ needed = $312.5 \times 12.01 = 3753 \text{ g} = 3.75 \text{ kg}^*$
 $m(\text{coal})$ needed = $3.75 \times 100/23.0 = 16.3 \text{ kg}^*$ (3 marks)
- b** **i** $\text{CH}_4(\text{g}) + 2\text{O}_2(\text{g}) \rightarrow \text{CO}_2(\text{g}) + 2\text{H}_2\text{O}(\text{g})$
(1 mark for correct reactants and products, 1 mark for correctly balancing equation) (2 marks)
- ii** 1.00 mol of CH_4 releases 890 kJ
So $1.00/16.05 \text{ mol}^*$ of CH_4 releases $890 \times 1.00/16.05 \text{ kJ} = 55.5 \text{ kJ}^*$ (2 marks)
- iii** Biodiesel or bioethanol (1 mark)
- c** 572 kJ are released by 2 mol of H_2 gas
So $3.00 \times 10^3 \text{ kJ}$ are released by $\frac{2 \times 3.00 \times 10^3}{572} = 10.5 \text{ mol}$ of H_2 gas^{*}
 $m(\text{H}_2)$ required = $m \times M = 10.5 \times 2.02 = 21.2 \text{ g}^*$ (2 marks)

Question 24

- a** Answer for the first row in the table: The structural formula and correct corresponding systematic name of one of the following: pentane, methylbutane or dimethylpropane.

Answer for the second row in the table: The structural formula and correct corresponding systematic name of one of the following: 1-chlorobutane, 2-chlorobutane, 1-chloro-2-methylpropane, 2-chloro-2-methylpropane.

The answer for the third row in the table: Accept any correctly drawn alkene or cycloalkane with correct corresponding name

(1 mark for each correct structural formula and systematic name) (6 marks)

- b** An unsaturated hydrocarbon is one that has one or more carbon-to-carbon double or triple bond. (1 mark)

- c i** Bromine or Br₂ (1 mark)

- ii** No visible reaction between bromine and a saturated compound*.

When bromine is added to an unsaturated compound, the brown colour of the bromine disappears*.

(2 marks)

- d** $\text{CH}_2\text{CHCH}_3(\text{g}) + \text{Cl}_2(\text{g}) \rightarrow \text{CH}_2\text{ClCHClCH}_3(\text{g})$

(1 mark for correct reactants, 1 mark for correct product. Accept correct equation using propyne.)

(2 marks)

End of questions